

LAB ACTIVITIES(11249A184) M.Lalitha Iyer

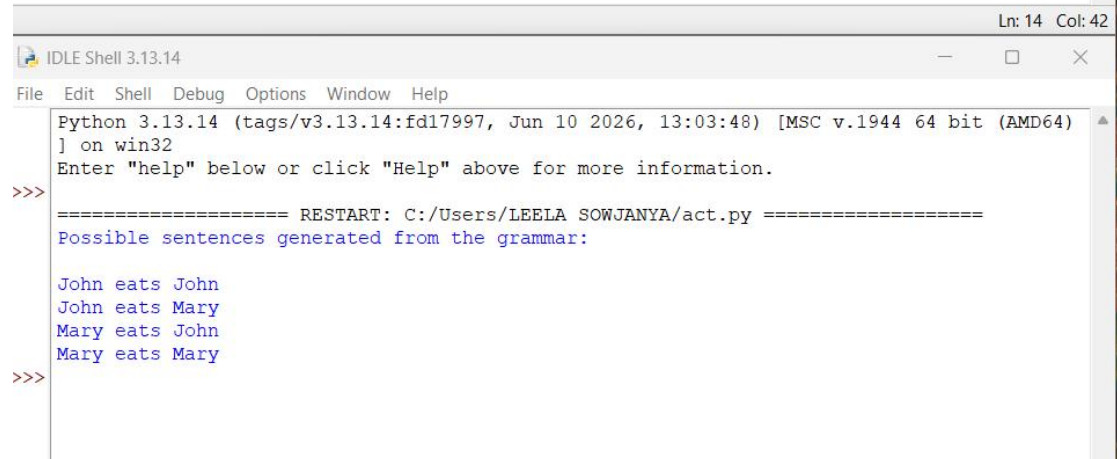
EXP-1(ACTIVITIES):-

```
# Grammar:
# S -> NP VP
# NP -> John | Mary
# VP -> eats NP

# Non-terminal options
NP = ["John", "Mary"]

print("Possible sentences generated from the grammar:\n")

# Generate all possible sentences
for subject in NP:
    for obj in NP:
        sentence = f"{subject} eats {obj}"
        print(sentence)
```



The screenshot shows an IDLE Shell window titled "IDLE Shell 3.13.14". The window has a menu bar with "File", "Edit", "Shell", "Debug", "Options", "Window", and "Help". The status bar at the top right indicates "Ln: 14 Col: 42". The main text area displays the output of a Python script. It starts with a prompt ">>>" followed by a message: "===== RESTART: C:/Users/LEELA SOWJANYA/act.py =====". Below this, it says "Possible sentences generated from the grammar:". Then, it lists four sentences: "John eats John", "John eats Mary", "Mary eats John", and "Mary eats Mary". The prompt ">>>" appears again at the bottom.

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/LEELA SOWJANYA/act.py =====
Possible sentences generated from the grammar:

John eats John
John eats Mary
Mary eats John
Mary eats Mary
>>>
```

```
# Grammar:|
# S -> NP VP
# NP -> John | Mary
# VP -> eats NP

# Define grammar rules
NP = ["John", "Mary"]
verb = "eats"

# Generate sentences
print("Generated Sentences:\n")

for subject in NP:
    for obj in NP:
        print(subject, verb, obj)
```

Ln: 1 Col: 10

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/LEELA SOWJANYA/act2.py =====
Generated Sentences:

John eats John
John eats Mary
Mary eats John
Mary eats Mary

>>>

```
File Edit Format Run Options Window Help
# Modified Grammar
VP = ["John", "Mary", "Student"]
NP = [
    ("eats", NP),      # VP -> eats NP
    ("studies", ["NLP"]) # VP -> studies NLP
]

print("Generated Sentences:\n")

for subject in NP:
    for verb, objects in VP:
        for obj in objects:
            print(subject, verb, obj)
```

Ln: 13 Col: 37

IDLE Shell 3.13.14

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```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/LEELA SOWJANYA/act3.py =====
Generated Sentences:

John eats John
John eats Mary
John eats Student
John studies NLP
Mary eats John
Mary eats Mary
Mary eats Student
Mary studies NLP
Student eats John
Student eats Mary
Student eats Student
Student studies NLP
>>>
```

```
File Edit Format Run Options Window Help

def generate_sentences(np, vp_rules):
    for subject in np:
        for verb, objects in vp_rules:
            for obj in objects:
                print(subject + " " + verb + " " + obj)

# ----- College Grammar -----
print("==== COLLEGE GRAMMAR ====")
college_np = ["Student", "Teacher", "Principal"]
college_vp = [
    ("studies", ["AI", "Python", "Mathematics"]),
    ("teaches", ["AI", "Python", "Mathematics"]),
    ("manages", ["College"])
]
generate_sentences(college_np, college_vp)

# ----- Hospital Grammar -----
print("\n==== HOSPITAL GRAMMAR ====")
hospital_np = ["Doctor", "Nurse", "Patient"]
hospital_vp = [

IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Reader issues AI Book
Reader issues Novel
Reader reads Python Book
Reader reads AI Book
Reader reads Novel

==== BANKING SYSTEM GRAMMAR ====
Customer deposits Cash
Customer withdraws Cash
Customer approves Loan
Banker deposits Cash
Banker withdraws Cash
Banker approves Loan
Manager deposits Cash
Manager withdraws Cash
Manager approves Loan
```

```
File Edit Format Run Options Window Help
# Function to generate sentences
def generate_sentences(np, vp_rules):
    for subject in np:
        for verb, objects in vp_rules:
            for obj in objects:
                print(subject + " " + verb + " " + obj)

# Online Shopping System Grammar
NP = ["Customer", "Seller", "Admin"]

VP = [
    ("buys", ["Laptop", "Mobile", "Book", "Shoes"]),
    ("sells", ["Laptop", "Mobile", "Book", "Shoes"]),
    ("adds to cart", ["Laptop", "Mobile", "Book", "Shoes"]),
    ("removes from cart", ["Laptop", "Mobile", "Book", "Shoes"]),
    ("places", ["Order"]),
    ("tracks", ["Order"]),
    ("manages", ["Inventory"])
]

print("==== ONLINE SHOPPING SYSTEM SENTENCES =====\n")
```

Ln: 23 Col: 0

IDLE Shell 3.13.14

```
File Edit Shell Debug Options Window Help
Customer buys Book
Customer buys Shoes
Customer sells Laptop
Customer sells Mobile
Customer sells Book
Customer sells Shoes
Customer adds to cart Laptop
Customer adds to cart Mobile
Customer adds to cart Book
Customer adds to cart Shoes
Customer removes from cart Laptop
Customer removes from cart Mobile
Customer removes from cart Book
Customer removes from cart Shoes
Customer places Order
Customer tracks Order
Customer manages Inventory
Seller buys Laptop
Seller buys Mobile
Seller buys Book
Seller buys Shoes
```

Ln: 159 Col: 0

EXP-2(ACTIVITIES):-

```
sentences = [
    "I like NLP",
    "I like Python",
    "I study NLP"
]

# ----- Word Frequency -----
word_freq = {}

for sentence in sentences:
    words = sentence.split()
    for word in words:
        word_freq[word] = word_freq.get(word, 0) + 1

print("Word Frequency:")
for word, count in word_freq.items():
    print(word, ":", count)

# ----- Bigram Frequency -----
bigram_freq = {}
```

Ln: 1

IDLE Shell 3.13.14

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Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/LEELA SOWJANYA/exp2act1.py =====

Word Frequency:

I : 3
like : 2
NLP : 2
Python : 1
study : 1

Bigram Frequency:

('I', 'like') : 2
('like', 'NLP') : 1
('like', 'Python') : 1
('I', 'study') : 1
('study', 'NLP') : 1

```
from collections import Counter

# Training corpus
corpus = "I like NLP I like Python I enjoy NLP"

# Tokenize
words = corpus.split()

# -----
# Unigram Probability
# -----
unigram_counts = Counter(words)
total_words = len(words)

print("Unigram Probabilities:")
for word, count in unigram_counts.items():
    print(f"P({word}) = {count}/{total_words} = {count/total_words:.3f}")

# -----
# Bigram Probability
# -----
```

Ln: 29 Col: 0

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/LEELA SOWJANYA/exp2act2.py =====

Unigram Probabilities:
P(I) = 3/9 = 0.333
P(like) = 2/9 = 0.222
P(NLP) = 2/9 = 0.222
P(Python) = 1/9 = 0.111
P(enjoy) = 1/9 = 0.111

Bigram Probabilities:
P(like|I) = 2/3 = 0.667
P(NLP|like) = 1/2 = 0.500
P(I|NLP) = 1/2 = 0.500
P(Python|like) = 1/2 = 0.500
P(I|Python) = 1/1 = 1.000
P(enjoy|I) = 1/3 = 0.333
P(NLP|enjoy) = 1/1 = 1.000

>>>

Ln: 20 Col: 0

ZIM - IND
In 7 hours

Search

```
exp2act3.py - C:/Users/LEELA SOWJANYA/exp2act3.py (3.13.14)
```

```
File Edit Format Run Options Window Help
# Count bigrams
bigrams = [(words[i], words[i+1]) for i in range(len(words)-1)]
bigram_counts = Counter(bigrams)

# Function to calculate sentence probability
def sentence_probability(sentence):
    tokens = sentence.split()
    probability = 1

    # Unigram probability for first word
    probability *= unigram_counts[tokens[0]] / len(words)

    # Bigram probabilities for remaining words
    for i in range(len(tokens)-1):
        bigram = (tokens[i], tokens[i+1])
        if bigram_counts[bigram] == 0:
            return 0
        probability *= bigram_counts[bigram] / unigram_counts[tokens[i]]
```

exp2act3.py - C:/Users/LEELA SOWJANYA/exp2act3.py (3.13.14)

File Edit Shell Debug Options Window Help

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/LEELA SOWJANYA/exp2act3.py =====

Probability of: I like NLP = 0.11111111111111111
Probability of: Python like I = 0

More probable sentence: I like NLP

>>>


```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/LEELA SOWJANYA/exp2act4.py =====
Bigram Probabilities with Laplace Smoothing:
P(I like|I) = 0.375
P(I like|Python) = 0.167
P(I like) = 0.143

Accuracy can be improved by:
1. Using larger training data
2. Applying smoothing techniques
3. Using trigrams or higher n-grams
4. Removing noise during preprocessing
5. Using neural language models
>>>

# Tokenization
words = corpus.split()

# Vocabulary
vocabulary = set(words)
vocab_size = len(vocabulary)

# Count unigrams
unigram_counts = Counter(words)

# Count bigrams
bigrams = [(words[i], words[i+1]) for i in range(len(words)-1)]
bigram_counts = Counter(bigrams)

# Bigram probability with Laplace smoothing
def bigram_probability(word1, word2):
    count_bigram = bigram_counts[word1, word2]
    count_word1 = unigram_counts[word1]
```

```
exp2act5.py - C:/Users/LEELA SOWJANYA/exp2act5.py (3.13.14)
File Edit Format Run Options Window Help

! Dataset
dataset = [
    "I love Python",
    "I love NLP",
    "I like Python",
    "I like Machine Learning",
    "Python is easy",
    "NLP is interesting"

! -----
! Grammar-Based Model
! -----
def grammar_model(sentence):
    grammar_rules = {
        "I love": ["Python", "NLP"],
        "I like": ["Python", "Machine"],
        "Python is": ["easy"],
        "NLP is": ["interesting"]
    }

Ln: 59 Col: 40
```

```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/LEELA SOWJANYA/exp2act5.py =====
Input: I love

Grammar-Based Model Output:
['Python', 'NLP']

Statistical Model Output:
[]
>>>
```



```
exp2act5.py - C:/Users/LEELA SOWJANYA/exp2act5.py (3.13.14)
File Edit Format Run Options Window Help
Dataset
dataset = [
    "I love Python",
    "I love NLP",
    "I like Python",
    "I like Machine Learning",
    "Python is easy",
    "NLP is interesting"

]

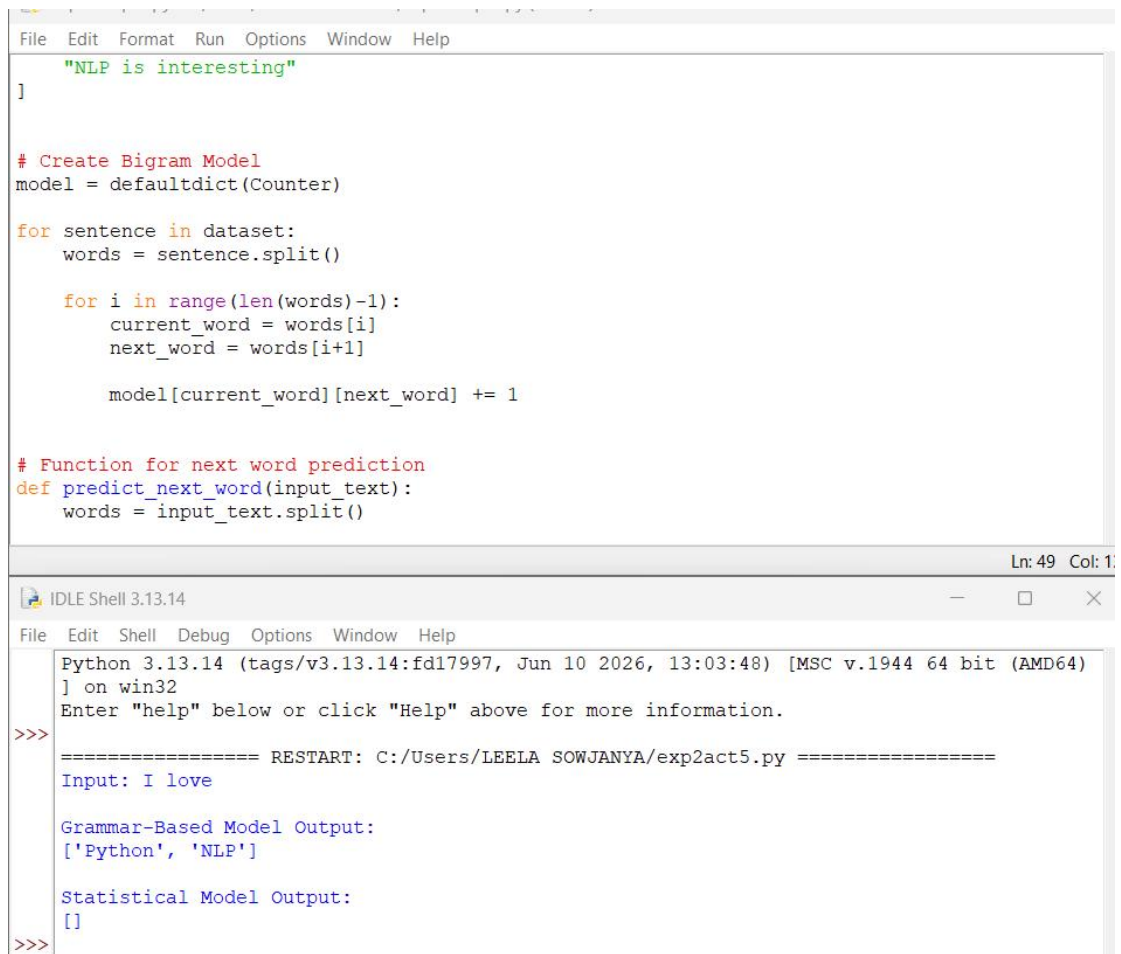
-----
Grammar-Based Model
-----
def grammar_model(sentence):
    grammar_rules = {
        "I love": ["Python", "NLP"],
        "I like": ["Python", "Machine"],
        "Python is": ["easy"],
        "NLP is": ["interesting"]
    }

Ln: 59 Col: 40

IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/LEELA SOWJANYA/exp2act5.py =====
Input: I love

Grammar-Based Model Output:
['Python', 'NLP']

Statistical Model Output:
[]
>>>
```



The image shows a Python IDE window with a menu bar (File, Edit, Format, Run, Options, Window, Help) and a code editor. The code defines a bigram model and a prediction function. Below the editor is an IDLE Shell window showing the execution of the code. The shell output includes a restart message, the input 'I love', and the outputs for both Grammar-Based and Statistical models.

```
File Edit Format Run Options Window Help

    "NLP is interesting"
]

# Create Bigram Model
model = defaultdict(Counter)

for sentence in dataset:
    words = sentence.split()

    for i in range(len(words)-1):
        current_word = words[i]
        next_word = words[i+1]

        model[current_word][next_word] += 1

# Function for next word prediction
def predict_next_word(input_text):
    words = input_text.split()

Ln: 49 Col: 1

IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> ===== RESTART: C:/Users/LEELA SOWJANYA/exp2act5.py =====
Input: I love

Grammar-Based Model Output:
['Python', 'NLP']

Statistical Model Output:
[]
>>>
```

EXP-3(ACTIVITIES):-

```
File Edit Format Run Options Window Help
import re

# Sample paragraph
text = """
Contact us at support@gmail.com or admin@yahoo.com.
Call us on 9876543210 or +91-9876543210.
Visit our website https://www.example.com or http://google.com
for more information.
"""

# Extract Email IDs
emails = re.findall(r'[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}', text)

# Extract Mobile Numbers
mobile_numbers = re.findall(r'(?:\+91[-\s]?[6-9]\d{9})', text)

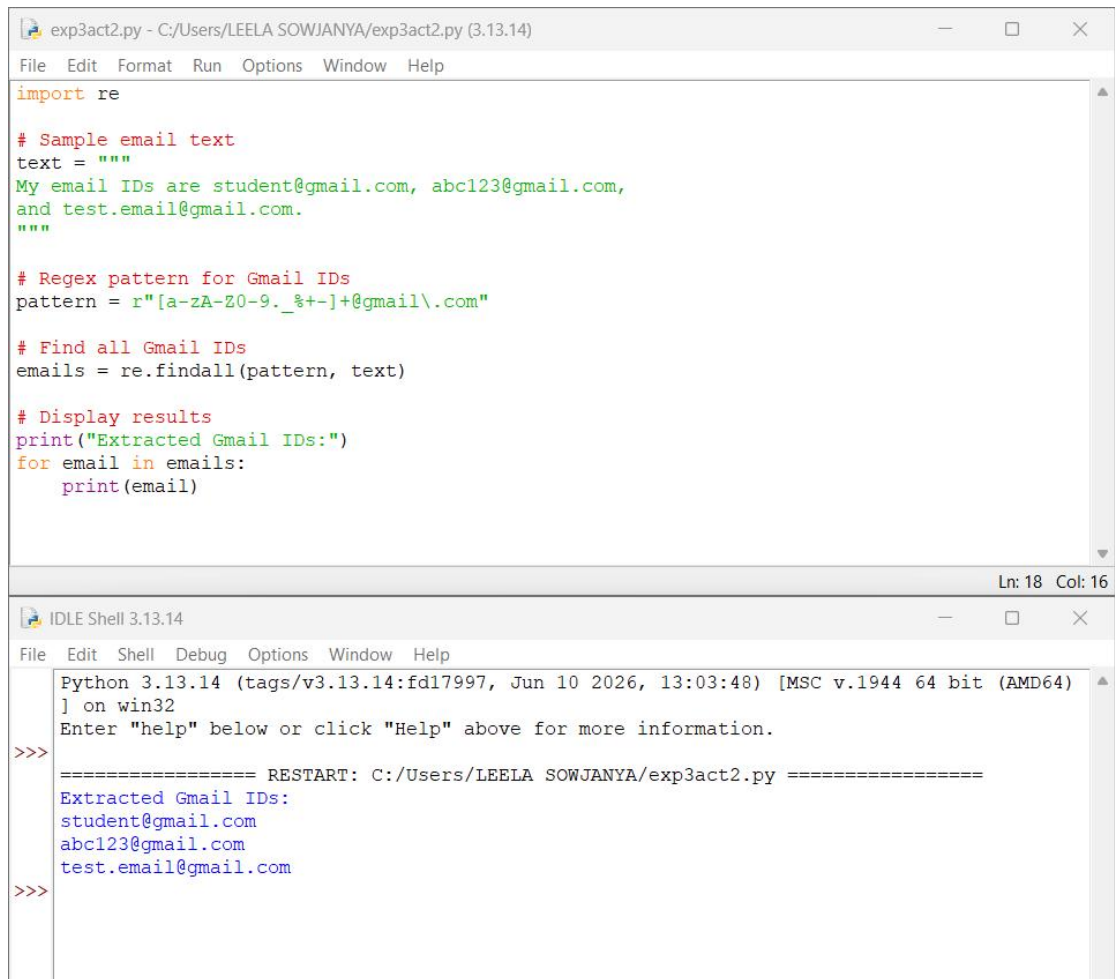
# Extract URLs
urls = re.findall(r'https?://[a-zA-Z0-9./_-]+', text)

# Display results
Ln: 32 Col: 14

IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/LEELA SOWJANYA/exp3act1.py =====
Email IDs:
support@gmail.com
admin@yahoo.com

Mobile Numbers:
9876543210
+91-9876543210

URLs:
https://www.example.com
http://google.com
>>>
```



The image shows a screenshot of a Python IDE with two windows. The top window, titled 'exp3act2.py - C:/Users/LEELA SOWJANYA/exp3act2.py (3.13.14)', contains a Python script. The script imports the 're' module, defines a sample email text string, and uses a regular expression pattern to find all Gmail IDs. The bottom window, titled 'IDLE Shell 3.13.14', shows the output of the script, which is the list of extracted email IDs: 'student@gmail.com', 'abc123@gmail.com', and 'test.email@gmail.com'.

```
exp3act2.py - C:/Users/LEELA SOWJANYA/exp3act2.py (3.13.14)
File Edit Format Run Options Window Help

import re

# Sample email text
text = """
My email IDs are student@gmail.com, abc123@gmail.com,
and test.email@gmail.com.
"""

# Regex pattern for Gmail IDs
pattern = r"[a-zA-Z0-9._%+-]+@gmail\.com"

# Find all Gmail IDs
emails = re.findall(pattern, text)

# Display results
print("Extracted Gmail IDs:")
for email in emails:
    print(email)

Ln: 18 Col: 16

IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/LEELA SOWJANYA/exp3act2.py =====
Extracted Gmail IDs:
student@gmail.com
abc123@gmail.com
test.email@gmail.com
>>>
```

```
import re

text = "Hello World"

result = re.findall(r'[a-z]', text)

print(result)
```

Ln: 7 Col: 1

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

```
>>>
===== RESTART: C:/Users/LEELA SOWJANYA/exp3act3.py =====
>>> ['e', 'l', 'l', 'o', 'o', 'r', 'l', 'd']
```

```

import re

# Sample resume text
resume = """
Name: Rahul Sharma
Email: rahul.sharma@gmail.com
Phone: +91-9876543210

Skills:
Python, Java, Machine Learning, Data Science, NLP
"""

# Extract Name
name = re.search(r"Name:\s*(.*)", resume)

# Extract Email
email = re.search(r"[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}", resume)

# Extract Phone Number
phone = re.search(r"(?:\+91[-\s]?)?[6-9]\d{9}", resume)

```

Ln: 1 Col: 9

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32

Enter "help" below or click "Help" above for more information.

```

>>> ===== RESTART: C:/Users/LEELA SOWJANYA/exp3act4.py =====
Name: Rahul Sharma
Email: rahul.sharma@gmail.com
Phone Number: +91-9876543210
Skills:
Python
Java
Machine Learning
Data Science
NLP
>>>

```

```

text = """
Aadhaar: 1234 5678 9012
PAN: ABCDE1234F
Vehicle: TN01AB1234
Date: 25-07-2026
Time: 10:30 AM
"""

# Aadhaar Number Pattern
aadhaar_pattern = r"\b\d{4}\s\d{4}\s\d{4}\b"

# PAN Number Pattern
pan_pattern = r"\b[A-Z]{5}[0-9]{4}[A-Z]\b"

# Vehicle Number Pattern
vehicle_pattern = r"\b[A-Z]{2}\d{2}[A-Z]{2}\d{4}\b"

# Date Pattern (DD-MM-YYYY)
date_pattern = r"\b\d{2}-\d{2}-\d{4}\b"

```

Ln: 41 Col

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
 Enter "help" below or click "Help" above for more information.

```

>>> ===== RESTART: C:/Users/LEELA SOWJANYA/exp3act5.py =====
Aadhaar Number:
['1234 5678 9012']

PAN Number:
['ABCDE1234F']

Vehicle Number:
['TN01AB1234']

Date:
['25-07-2026']

Time:
['AM']

```



```

# Extract Education
# -----
education = re.search(
    r"Education:\s*(.*?)\n\n",
    resume,
    re.DOTALL
)

# -----
# Extract Experience
# -----
experience = re.search(
    r"Experience:\s*(.*)",
    resume,
    re.DOTALL
)

# Display Extracted Information

print("----- Resume Information -----")

```

Ln: 92 Col: 26

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/LEELA SOWJANYA/exp3minipro.py =====
----- Resume Information -----

Name:
Ananya Kumar

Email:
ananya.kumar@gmail.com

Phone:
+91-9876543210

Skills:
- Python
- Java
- Machine Learning
- NLP

EXP-4(ACTIVITIES):-

```
# DFA for Regular Expression: ab*

def dfa_abstar(input_string):

    # Initial state
    state = "q0"

    # Process each character
    for symbol in input_string:

        if state == "q0":
            if symbol == "a":
                state = "q1"
            else:
                state = "dead"

        elif state == "q1":
            if symbol == "b":
                state = "q1"
            else:
                state = "dead"
```

Ln: 37 Col: 24

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

```
>>> ===== RESTART: C:/Users/LEELA SOWJANYA/exp4act1.py =====
Enter a string: 2
String: 2
Result: Rejected
>>>
```

```

# DFA for accepting language ab*
def dfa_abstar(string):
    state = "q0"
    for symbol in string:
        if state == "q0":
            if symbol == "a":
                state = "q1"
            else:
                state = "dead"

        elif state == "q1":
            if symbol == "b":
                state = "q1"
            else:
                state = "dead"

    else:

```

Ln: 35 Col: 32

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

```

>>> ===== RESTART: C:/Users/LEELA SOWJANYA/exp4act1.py =====
Enter a string: 2
String: 2
Result: Rejected

```

```

>>> ===== RESTART: C:/Users/LEELA SOWJANYA/exp4act2.py =====
a : Accepted
ab : Accepted
abb : Accepted
abbb : Accepted

```

```

>>>

```

```

# Incorrect DFA Transition Table

incorrect_table = {
    "q0": {"a": "q0", "b": "q1"},
    "q1": {"a": "q0", "b": "q1"}
}

print("Incorrect Transition Table:")
print("State\t a\t b")

for state, transitions in incorrect_table.items():
    print(state, "\t", transitions["a"], "\t", transitions["b"])

# Debugging Errors

print("\nErrors Found:")
print("1. q0 on input 'a' should move to q1, not q0.")
print("2. q0 on input 'b' should move to Dead state.")
print("3. q1 on input 'a' should move to Dead state.")

```

Ln: 35 Col: 64

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/LEELA SOWJANYA/Desktop/exp4act3.py =====

Incorrect Transition Table:

State	a	b
q0	q0	q1
q1	q0	q1

Errors Found:

1. q0 on input 'a' should move to q1, not q0.
2. q0 on input 'b' should move to Dead state.
3. q1 on input 'a' should move to Dead state.

Correct Transition Table:

State	a	b
q0	q1	Dead
q1	Dead	q1
Dead	Dead	Dead

```

# DFA Simulation Game

# Transition table
dfa = {
    "q0": {"a": "q1", "b": "q2"},
    "q1": {"a": "q1", "b": "q2"},
    "q2": {"a": "q2", "b": "q2"}
}

# Start state
current_state = "q0"

# Input symbols entered by student
input_string = input("Enter input symbols: ")

print("\nDFA Simulation:")
print("Starting State:", current_state)

# Process symbols one by one
for symbol in input_string:

```

Ln: 35 Col: 38

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/LEELA SOWJANYA/exp4act4.py =====
Enter input symbols: 4

DFA Simulation:
Starting State: q0
Invalid Symbol

Final State: q0
>>>

```

```

def divisible_by_2(binary):
    if binary[-1] == '0':
        return "Accepted (Divisible by 2)"
    else:
        return "Rejected (Not divisible by 2)"

number = input("Enter binary number: ")
print(divisible_by_2(number))

```

Ln: 11 Col: 29

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

```

>>>
===== RESTART: C:/Users/LEELA SOWJANYA/exp4act5.py =====
Enter binary number: 110
Accepted (Divisible by 2)
>>>

```

```

# DFA Simulator

def dfa_simulator():

    # DFA Transition Table
    transitions = {
        "q0": {"a": "q1", "b": "dead"},
        "q1": {"b": "q1", "a": "dead"},
        "dead": {"a": "dead", "b": "dead"}
    }

    start_state = "q0"
    final_state = "q1"

    # User input
    string = input("Enter input string: ")

    current_state = start_state

    # DFA processing
    for symbol in string:

```

Ln: 39 Col: 1

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

```

>>>
===== RESTART: C:/Users/LEELA SOWJANYA/exp4minipro.py =====
Enter input string: 6
Input: 6 -> State: dead

String Rejected
>>>

```